

FINAL EXPLANATION: CHEMISTRY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CHEMISTRY GRADE 10 – SUB-STRAND 3.1 ACIDS AND BASES: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Remember the eight cups at the front of the class in Lesson 1: lemon juice, black tea (chai), cow's milk, Stoney Tangawizi, tap water, baking soda solution, liquid handwash soap and dilute Jik bleach. Five drops of universal indicator turned them into a rainbow. Our Driving Question was: "What invisible properties of everyday Kenyan liquids cause them to turn such different colours, and why does it matter for our health, our farms, and our environment?" In the space under each prompt, write a full scientific explanation in your own words. Use the Arrhenius definitions, ions, the pH scale, indicator behaviour and the chemical properties of acids and bases that you learned in Lessons 1 to 10. Use evidence from our experiments and give chemical equations where asked. Write in full sentences.

Part 1: Sorting the cups into groups

Look back at the eight cups from Lesson 1. (a) Sort all eight cups into three groups (acidic, neutral, basic) and state the universal indicator colour you observed for each group. Make sure every cup is placed in a group. (b) Describe how red litmus, blue litmus, phenolphthalein and methyl orange would each behave in an acid and in a base. (c) Explain why we discuss taste and feel as properties of acids and bases but never use them to test substances in the laboratory.

(a) Acidic group: lemon juice, Stoney Tangawizi and black tea (chai) turned red or orange with universal indicator. Neutral group: tap water turned a stable green, and cow's milk, which is neutral or only very slightly acidic, gave a yellow-green colour next to green on the indicator range, so it belongs in the neutral group. Basic group: baking soda solution, liquid handwash soap and dilute Jik bleach turned blue or purple. (b) In an acid: blue litmus turns red (red litmus stays red), phenolphthalein stays colourless, methyl orange turns red. In a base: red litmus turns blue (blue litmus stays blue), phenolphthalein turns pink/magenta, methyl orange turns yellow. A neutral substance causes no litmus change, leaves phenolphthalein colourless and leaves methyl orange orange-yellow. (c) Acids taste sour and bases taste bitter and feel slippery, but many laboratory chemicals are corrosive or poisonous, so tasting or touching them is dangerous. Indicators give a safe, reliable and reproducible way to classify a liquid, so we never taste substances in the laboratory.

Part 2: What is inside an acid or a base?

Explain at the particle level why the acidic cups and the basic cups gave different colours. In your answer: (a) state the

(a) An Arrhenius acid produces H^+ ions in aqueous solution. An Arrhenius base produces OH^- ions in aqueous solution. (b) $HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$. $H_2SO_4(aq) \rightarrow 2H^+(aq) + SO_4^{2-}(aq)$ (diprotic; HCl is monoprotic). $NaOH(aq) \rightarrow Na^+(aq) + OH^-(aq)$. $Ca(OH)_2(aq) \rightarrow Ca^{2+}(aq) + 2OH^-(aq)$. (c) Lemon juice and Stoney Tangawizi contain dissolved acids that release H^+ ions, so the cups have a high H^+ concentration and universal indicator turns red or orange. Liquid soap and Jik bleach are basic solutions with

Arrhenius definition of an acid and of a base; (b) write ionisation equations for HCl, H ₂ SO ₄ , NaOH and Ca(OH) ₂ ; (c) explain how the particles you named link to the colours of lemon juice and liquid soap with universal indicator, and to what happens in tap water.	a high OH ⁻ concentration, so universal indicator turns blue or purple. Tap water has equal amounts of H ⁺ and OH ⁻ , so it is neutral and gives green. The colours are not random: they reveal which ion is present in greater amount.
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Part 3: Why the shades differ – pH, strength and ionisation

The cups did not all show the same shade within the red group or within the blue-purple group. (a) Explain what the pH scale measures and how pH numbers relate to the indicator colours and to H⁺ concentration. (b) In Lesson 9 hydrochloric acid gave pH 1 while ethanoic acid in vinegar gave about pH 3, and sodium hydroxide gave pH 13 while ammonia solution gave about pH 11. Suggest what property of the particles in these solutions could account for the differences, and show how the equations for a strong and a weak acid differ. (c) Use your ideas to explain why lemon juice (red) and Stoney Tangawizi (orange-red) differ in shade, and why the basic cups differ in shade from each other.	(a) pH is a scale from 0 to 14 that measures the relative concentration of H⁺ ions. pH below 7 is acidic, 7 is neutral and above 7 is basic. Each step is about a tenfold change in H⁺ concentration. Lower pH means more H⁺ and a redder colour with universal indicator; higher pH means more OH⁻ and a bluer or purple colour. Lemon juice is about pH 2, tap water about pH 7 and Jik about pH 12 to 13. (b) The difference lies in the degree of ionisation. Strong acids and bases such as HCl and NaOH ionise completely, giving many H⁺ or OH⁻ ions and pH values far from 7. Weak acids and bases such as ethanoic acid and ammonia ionise only partially, so fewer ions are produced and the pH is closer to 7. A strong acid uses a one-directional arrow: $\text{HCl} \rightarrow \text{H}^+ + \text{Cl}^-$. A weak acid uses a reversible arrow: $\text{CH}_3\text{COOH} \rightleftharpoons \text{H}^+ + \text{CH}_3\text{COO}^-$. (c) Lemon juice (citric acid, a weak acid, pH about 2) gave red, and Stoney Tangawizi (carbonic acid, a weak acid, pH about 3 to 4) gave orange-red: the cup with the lower pH has the higher H⁺ concentration and the redder colour. In the same way, the basic cups differ in shade because they differ in OH⁻ concentration. Jik bleach, a strongly basic solution at pH about 12 to 13, gave the deepest purple, while baking soda solution and liquid soap, with a lower OH⁻ concentration and lower pH, gave blue or blue-violet. Shade therefore depends on how many H⁺ or OH⁻ ions are present, which depends on both the concentration of the substance and how much of it ionises.
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Part 4: What acids do – chemical properties

Acids are not only detected by colour; they also react. (a) Name the four types of substance that acids react with in the lessons, and for each	(a) Acid + metal (e.g. zinc or magnesium) → salt + hydrogen gas; evidence is bubbling and a 'pop' with a lit splint. Acid + metal oxide (e.g. copper oxide) → salt + water; evidence is the black solid dissolving to give a blue solution. Acid + metal carbonate or hydrogen carbonate (marble chips, baking soda) → salt + water + carbon dioxide; evidence is fizzing and limewater turning milky. Acid + base/alkali (neutralisation) → salt + water; evidence is phenolphthalein turning from colourless to pink at the endpoint when NaOH is added to HCl. (b) $\text{Zn(s)} + 2\text{HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$. $2\text{HCl(aq)} + \text{CaCO}_3\text{(s)} \rightarrow \text{CaCl}_2\text{(aq)} + \text{H}_2\text{O(l)} + \text{CO}_2\text{(g)}$. Net
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give the products and the test or observation that provides the evidence. (b) Write one balanced equation for acid + metal, one for acid + carbonate, and the net ionic equation for neutralisation. (c) Explain what these reactions have in common in terms of the Arrhenius model.	ionic equation for neutralisation: $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$. (c) In every reaction the H^+ ion from the acid is the active particle. It is the same particle that makes universal indicator turn red, so the cups' colours and the reactions both come from H^+ ions.
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Part 5: Why it matters – health, farms and environment

Answer the second half of the Driving Question. Using examples from Kenya, explain why the acid or base properties of liquids matter for (a) health, (b) farming and (c) the environment or industry. In each case name the substance, say whether it is an acid or a base, and explain the chemistry involved. Finally, write a short concluding paragraph that answers the whole Driving Question, referring to all eight cups.	(a) Health: stomach acid (HCl, pH about 1 to 2) helps digest foods such as ugali and githeri. When there is too much, antacids containing bases such as magnesium hydroxide or sodium bicarbonate neutralise the excess H^+ ions (for example $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$ for magnesium hydroxide), relieving discomfort. (b) Farming: in Kericho, tea soil with pH 4.2 is too acidic and harms the plants, while pH 5.5 to 6.0 supports good growth. Farmers apply lime to neutralise the acid and raise the pH. If the lime is calcium hydroxide, a strong base, the OH^- ions react with H^+ ($\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$); if it is calcium carbonate, the carbonate ions react with H^+ to give water and carbon dioxide ($2\text{H}^+ + \text{CO}_3^{2-} \rightarrow \text{H}_2\text{O} + \text{CO}_2$). Lime is also used on acidic soils in the Rift Valley. (c) Environment and industry: sulphuric acid is used in car batteries and can corrode metals and carbonate materials; sodium hydroxide is used in soap making at Kisumu, and wastes that are too acidic or basic can change the pH of rivers such as the Nairobi River, so effluent pH must be checked. Conclusion: the eight cups turned different colours because each liquid has a different concentration of H^+ or OH^- ions. Acidic liquids such as lemon juice, Stoney Tangawizi and chai have more H^+ and are red to orange; tap water and cow's milk are neutral or very nearly so and are green to yellow-green; baking soda, liquid soap and Jik have more OH^- and are blue to purple. How strong the acid or base is, shown by its degree of ionisation and pH, explains the different shades. These same ions control how substances react, so measuring and controlling pH matters for our health, farms and environment.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Classification and use of evidence	Correctly sorts all eight cups and accurately uses indicator colours and observations from the experiments to support every claim.	Sorts most cups correctly and uses some indicator evidence, with minor errors or gaps.	Sorts few cups correctly or gives colours without linking them to acid, base or neutral.
Particle-level explanation (Arrhenius model)	States both Arrhenius definitions correctly, writes correct ionisation equations, and links H^+ and OH^- to indicator colours.	States the definitions and writes most equations correctly, but the link to the colours is incomplete.	Gives incomplete or incorrect definitions or equations, with little reference to ions.
pH, strength and ionisation	Explains the pH scale and the tenfold change, and clearly distinguishes strong	Explains pH and the strong/weak difference with some imprecision.	Confuses strength with concentration or pH values, or omits ionisation.

	from weak using degree of ionisation and correct arrows.		
Chemical properties and equations	Gives all four reaction types with correct products, tests and balanced and ionic equations, and identifies H ⁺ as the common active particle.	Gives most reaction types, with minor errors in equations or tests.	Gives few reactions or has several incorrect products or equations.
Kenyan applications and final conclusion	Gives accurate Kenyan examples for health, farming and environment with correct chemistry, and a clear conclusion answering the Driving Question.	Gives relevant examples with partial chemistry and a conclusion that answers most of the Driving Question.	Gives vague or incorrect examples and a conclusion that does not address the Driving Question.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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